

The syntax for the 'Hello World' program in Scala is:

```
object HelloWorld extends App {
  println("Hello, World!")
}
```

The advantages of Scala

- Scala enables developers to write simple and straightforward text, as it requires two-thirds less code than Java and that code is more flexible. This, in turn, makes the code human-readable and easier to understand.
- Scala enables quick implementation and enhanced performance, as it reduces various thread-safety concerns and treats functions on first priority.
- Scala solves various concurrency issues because it is well equipped with the Actor library.
- Scala has the fastest growing ecosystem compared to any other language in the current scenario, as the IDE tools, testing tools, documentation and even the libraries are improving, while its capabilities are getting enhanced.

Scala's disadvantages

- Scala is right now very hard to learn as the language is completely different from pure Java and the programming environment is also different, requiring more professional developers to understand and start building up the logic and developing bug-free code.
- Even today, there are very few projects written purely in Scala. In a nutshell, we see projects with a hybrid composition of both Java and Scala, and mixing up these two is somewhat time consuming and slow. And testing becomes a pain.
- Because of limited backward compatibility, Scala is not compatible with previous versions, which creates problems when the development time period is short. So, Scala is not recommended for projects with time constraints.

6. Dart

Dart is a class-based, single inheritance, object-oriented-general-purpose programming language originally developed by Google and later approved as a standard by ECMA (ECMA-408). Being an open source programming language enables Dart developers to build more complex, high performance and scalable-rich apps for the modern Web. Dart comes bundled with a Web Component Library, which contains Web code comprising HTML and JavaScript, which can be used in different pages or projects.

Dart has tools such as:

- **Dartboard:** Enables the developers to write and run Dart code in the Web browser.

- **Dart Editor:** Enables the developers to create, modify and run Dart apps.
- **SDK:** Contains command-line tools such as a Dart-to-JavaScript compiler, a Dart virtual machine, etc.
- **Dartium:** Contains a built-in Dart VM. The latest stable release of Dart is 1.18. The syntax for the 'Hello World' program in Dart is:

```
void main() {
  print('Hello, World!');
}
```

The advantages of Dart

- Dart is regarded as more than just a programming language, providing a good platform for Web developers and a new feature set that includes optional typing and isolates.
- Dart core libraries provide functionality including collections, dates, math, HTML bindings, server-side I/O like sockets and even JSON.
- Dart's VM has been built from scratch. It can run on the command line for server-side applications, and can be embedded into browsers for client-side applications.
- Dart has other useful features like mixins, implicit interfaces, lexical closures, lexical this, named constructors, string interpolation, online functions and noSuchMethod.

Dart's disadvantages

- Dart's SDK doesn't provide access to SQL based databases at the server level. Third party packages come to the rescue.
- Dart is currently not very successful in generating consumable JavaScript. The entire app build in Dart has to be built in JavaScript at once, for now.
- Dart is not entirely interoperable with JavaScript right now as the Dart JavaScript library is not stable. It is currently in the development stage.

7. Clojure

Clojure is regarded as a dynamic, general-purpose programming language that combines the approachability and interactive development of a scripting language with efficient and robust infrastructure for multi-threaded programming. Clojure is closely related to Lisp, which was created by Rich Hickey. It operates on the Java VM and is integrated with Java. It fully supports calling Java code from Clojure and vice-versa. Clojure supports functions as first class objects, a read-eval-print loop (REPL) and a macro system. It encourages the use of first-class and high-order functions with values, and comes with its own set of efficient immutable data structures. Clojure offers innovative solutions to the challenges inherent in